

MATERIALS SYMPOSIUM • POSTER 214

# A polymer that stitches its own cracks back together

Reversible bonds give a structural plastic that recovers 90% of its strength after fracture.



# **We built a load-bearing plastic that heals a clean break in ten minutes at room temperature.**

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Most self-healing polymers are soft gels or need an oven to recover. Ours is a rigid, structural material that re-bonds across a fracture surface at room temperature — recovering 90% of its original strength without any external trigger.



# The problem: structural plastics fail at invisible micro-cracks.

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In load-bearing components, fracture starts as micro-cracks far below the surface, invisible until the part snaps. A material that closes those cracks as they form would extend service life dramatically — but the chemistry that allows healing usually makes the material too soft to bear load.

- Micro-cracks nucleate failure long before anything is visible.
- Healing chemistry and structural stiffness normally trade off.



# Our idea: two bond types — strong ones for strength, weak ones for healing.

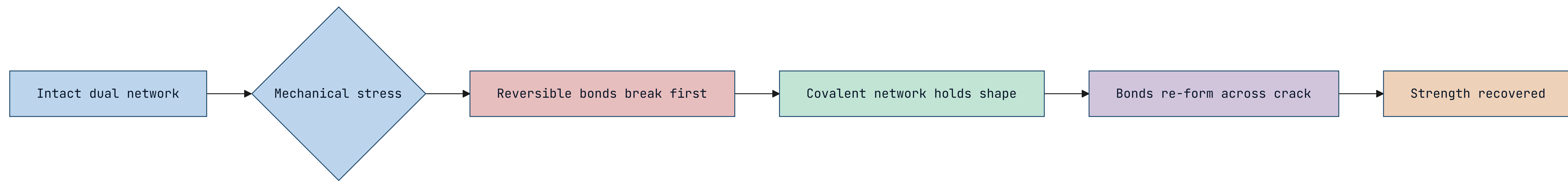
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We weave two networks into one polymer. Permanent covalent bonds carry the load and give the material its rigidity. Reversible hydrogen bonds break first under stress and re-form across a crack face — sacrificial bonds that heal while the strong network holds the shape.



# How the dual network absorbs a crack and closes it.

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# The mechanism: sacrificial bonds dissipate energy, then reconnect.

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When stress hits, the weak hydrogen bonds rupture and absorb the energy that would otherwise propagate a crack. Because the strong network keeps the two faces aligned, the weak bonds find their partners again and re-form. The damage becomes reversible rather than permanent.

- Weak bonds dissipate fracture energy as they break.
- The strong network keeps crack faces aligned for re-bonding.



MECHANICAL TESTING • N = 30 SPECIMENS

# What the dual-network polymer recovers after a clean fracture.

*Tensile bars fractured, rejoined for 10 min at 23°C, then re-tested.*

**90%**

STRENGTH RECOVERED

**10 min**

TO HEAL AT 23°C

**50×**

HEAL CYCLES BEFORE FATIGUE

**38 MPa**

TENSILE STRENGTH



# It heals many times over — not just once.

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We fractured and healed the same specimen fifty times before recovery dropped below 80%. Each cycle re-forms the reversible bonds with little degradation of the permanent network. The material treats damage as routine, not terminal.

- Recovery stays above 80% through 50 fracture cycles.
- The covalent backbone survives repeated healing intact.



# The trade-off we are still negotiating: heat softens the healing bonds.

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Above 60°C the reversible network loosens and the material creeps under sustained load. For room-temperature applications this is invisible, but high-temperature use needs a stronger reversible chemistry. That is the next dial to turn.

- Service ceiling is currently about 60°C.
- A higher-temperature reversible bond is the next target.



# Where this goes: components that report and repair their own fatigue.

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We are blending the healing chemistry with a strain-responsive dye so a part changes colour where it has healed — a visual fatigue log. The aim is structural components that both repair micro-damage and show their repair history at a glance.

- A strain-responsive dye marks healed regions.
- The part becomes its own maintenance record.



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**Build the weakness in on purpose,  
and the material learns to heal.**